

CABBY

Online Gaming Portal

Product Requirements Document

Complete specification for the CABBY online gaming portal

1. Project overview

CABBY is a browser-based online gaming portal where users can discover and play free HTML5 games without downloading anything. Users visit the site, browse a grid of games sorted by category, click one, and play it directly inside the browser via an iframe.

The portal includes four engagement features beyond standard browsing: a daily missions system that gives all users the same 3 missions per calendar day, a streak counter that rewards consecutive days of completing all 3 missions, a pause bookmarks system that lets users save where they left off in any game with a personal note, and an automatic highest score tracker that records each user's best score per game.

The portal is a frontend-only project. There is no backend server, no database, no Firebase, and no API keys. All user data is stored in the browser using localStorage. Games are not built by the developer. They are third-party HTML5 games embedded via iframe.

1.1 Project at a glance

Project name	CABBY: Online Gaming Portal
Project type	Browser-based online gaming portal
Platform	Browser, desktop and tablet
Tech stack	HTML, CSS, JavaScript. React acceptable but not required
Backend	None. No server, no database, no API keys
Data storage	Browser localStorage only
Engagement features	Daily missions, streak counter, pause bookmarks, best score tracker
Hosting	GitHub Pages or Vercel free tier
Deliverables	Source code on GitHub, live deployed link, README

2. Assets provided

The following files accompany this document. Do not replace or modify these assets.

File	Description	Usage
logo.png	CABBY brand logo. Yellow steering-wheel icon with CABBY text on a dark background	Place in the top-left corner of the navigation bar on all pages
reference_image.png	Portal layout reference showing the navigation bar, sidebar, missions section, and game card grid	Use as the visual target for overall layout, card design, and mission card placement

3. Deliverables

The freelancer must deliver three items at the end of the project.

#	Deliverable	Format	Notes
1	Source code	Public GitHub repository	All source files plus README.md with setup instructions
2	Live site	Public URL	Deployed on GitHub Pages or Vercel free tier

#	Deliverable	Format	Notes
3	README.md	Markdown file in repo root	Must include tech stack, setup steps, and live link

Ownership

All source code, assets created during the project, and deployment rights transfer to the client on final payment. The freelancer may not resell or reuse any part of this project.

4. Website pages

The portal has exactly four pages. All pages share a top navigation bar that includes the streak counter. The Home, Leaderboard, and Profile pages also share a left sidebar.

#	Page	Purpose
4.1	Home page	Today's missions, browse and search games, filter by category
4.2	Game page	Play the selected game inside an iframe. Save scores, bookmarks, view personal best
4.3	Leaderboard page	Tab switcher between All Scores and Your Best. Clear all option
4.4	Profile page	Display name, four stat cards, favourites, bookmarks, reset

4.1 Home page

The main landing page. Contains: top navigation bar with streak counter, left sidebar, Today's Missions section, and the game card grid.

Top navigation bar

Element	Position	Behaviour
Logo and brand text	Left	Logo image at 28x28px. Text CABBY in #F5C842, 16px, weight 500
Search bar	Centre	Placeholder "Find game or genre". Filters grid in real time
Streak counter	Right section	Flame icon in #EF9F27 followed by current streak number in #F5C842
Credits icon	Right section	Bolt icon in #1D9E75 next to 0. Static display
Notification bell	Right section	Bell icon with red badge. Static placeholder
User icon	Right section	Person icon in #999
Log in button	Far right	Background #1D9E75. Opens modal to set display name

Today's Missions section

Positioned above the game grid. Contains a heading "Today's Missions" and exactly 3 mission cards in a row. Each mission card shows the mission text and a completion indicator. Missions auto-complete. Completed cards use a green-tinted background (#1D3328) and show a green checkmark.

Left sidebar

A 48px wide vertical strip. Contains icon buttons stacked vertically, each 34x34px. Categories: All Games, Trending, Favourites, then a dot separator, then Racing, Puzzle, Action, Runner, Sports, New, Settings. Active icon uses background #3A3A3A and colour #F5C842.

Game card grid

Minimum 9 game cards on load. Grid uses 3 columns at 1024px and wider, 2 columns at 768px to 1023px. Gap between cards: 10px.

Card element	Specification
Aspect ratio	16:10
Background	Dark gradient, unique per genre

Card element	Specification
Category icon	Centred, 32px, colour matching the genre
Game name	Below icon, 12px, #DDDDDD, weight 500
Rating badge	Bottom-left. Background #1D9E75, number in white 10px weight 500
Bolt icon	Top-left, #1D9E75
Info button	Top-right, 18px circle with info icon
Age tag	Bottom-right, "16+" in #999 at 9px. Only on applicable games

4.2 Game page

Opens when a user clicks a game card. Contains the top nav bar with streak counter, no sidebar.

Element	Specification
Back button	Left arrow icon and "Back to games" text. Returns to Home page
Game title	Displayed above the iframe. 18px, white, weight 500
Best score display	Above the iframe in the format "Your best: N" where N is read from <code>cabby_best_scores</code> . Shows "Your best: none" when no best exists for the game
Iframe	Fills remaining width, minimum height 400px. Loads the game URL. Sandbox: <code>allow-scripts allow-same-origin</code>
Save Score input	Below iframe. Number input. Placeholder "Enter your score"
Save Score button	Background #1D9E75. Saves {game, score, date} to <code>cabby_scores</code> . Also updates <code>cabby_best_scores</code> for the game ID only if the new score exceeds the stored best
Save Bookmark button	Opens an inline text input that accepts up to 80 characters with a Save action. Stores {id, gameId, gameName, note, date, lastScore} to <code>cabby_bookmarks</code>
Add to Favourites button	Heart icon toggle. Adds the game ID to <code>cabby_favourites</code> on first click, removes it on the next click
Error fallback	If iframe fails to load: display "Game could not be loaded. Try another one."

4.3 Leaderboard page

Displays saved scores with two views. Contains the top nav bar with streak counter and left sidebar.

Element	Specification
Page title	Leaderboard with trophy icon
Tab switcher	Two tabs labelled "All Scores" and "Your Best". Switching tabs changes the visible table without reloading the page. The active tab is visually distinct
All Scores table	4 columns: Rank, Game Name, Score, Date. Sorted by Score descending. Data from <code>cabby_scores</code>
Your Best table	4 columns: Rank, Game Name, Highest Score, Date achieved. One row per game with a recorded best score. Sorted by Highest Score descending. Data from <code>cabby_best_scores</code>
Empty state	When the active tab has no data, show "No scores yet" or "No best scores yet"
Clear All Scores button	Background #E24B4A. Confirmation prompt before deleting <code>cabby_scores</code> . Best scores are preserved

4.4 Profile page

Shows the user's local profile. Contains the top nav bar with streak counter and left sidebar.

Element	Specification
Avatar	44px circle. User initials in white on #1D9E75 background. If no name: "?"
Display name	18px, weight 500, colour #FFFFFF. If not set: "Guest"
Edit Name button	Opens inline text input. Saves to cabby_username
Stat: Games Played	Count of unique game names in cabby_scores
Stat: Total Scores Saved	Total entries in cabby_scores array
Stat: Favourites	Count of items in cabby_favourites array
Stat: Best Streak	Value of cabby_streak.best, with current value shown beneath
Favourites list	All favourited game names with a remove button per item
Bookmarks list	All saved bookmarks from cabby_bookmarks showing game name, note, and date. Each item has a delete button. Clicking the game name navigates to the Game page
Reset Profile button	Background #E24B4A. Confirmation prompt. Clears all cabby_ keys

5. Daily missions and streaks

CABBY uses a daily missions system to encourage repeat visits. Each calendar day, all users see the same 3 missions chosen deterministically from a fixed pool of 10. Completing all 3 missions in a single day increments the user's streak. Missing a day resets the current streak.

5.1 Mission pool

#	Mission template
1	Play any game for 2 minutes total
2	Score above 500 in any game
3	Try a game you have never played before
4	Play games from 3 different categories
5	Add a game to favourites
6	Save 3 scores in one day
7	Set or change your display name
8	Play a game for 5 minutes total
9	Score above 1000 in any game
10	Visit the leaderboard page

5.2 Mission selection and streak logic

Rule	Specification
Missions per day	Exactly 3
Determinism	Same 3 missions for every user on the same calendar date
Selection formula	Any deterministic function of date yielding 3 distinct missions from the pool
Reset time	Local midnight in the user's browser timezone
Auto-complete	Mission checks fire whenever the relevant user action occurs
Mission card height	Approximately 80px
Mission card border radius	8px
Mission card grid	3 columns at viewport widths above 768px, 10px gap
Completed visual	Background #1D3328, green checkmark icon in #1D9E75
Streak increment	When all 3 of today's missions are complete, <code>cabby_streak.current</code> increases by 1
Best streak update	<code>cabby_streak.best = max(best, current)</code> on every increment, never reset
Streak reset	On page load, if today's date is more than 1 day after <code>lastCompletedDate</code> , current resets to 0
Nav display	Flame icon in #EF9F27 + current value in #F5C842, on every page

6. Pause bookmarks

Pause bookmarks let users mark a moment in any game and return to it later with a personal note. Bookmarks are stored locally and managed from the Profile page.

Rule	Specification
Trigger	Save Bookmark button on the Game page, placed next to the Save Score button
Input	Clicking opens an inline text input that accepts up to 80 characters with a Save action
Storage key	cabby_bookmarks (JSON array of bookmark objects)
Bookmark fields	id (unique string), gameId, gameName, note, date (YYYY-MM-DD), lastScore (the user's last entered score for that game, or 0)
Display location	Bookmarks section on the Profile page
List item content	Game name, note, date, and a delete button
Delete action	Removes that bookmark object from cabby_bookmarks
Navigation	Clicking the game name of any bookmark navigates to the Game page for that bookmarked game

7. Highest score tracker

Each user's highest score per game is tracked automatically, separate from the score history. Best scores are displayed on the Game page and on a dedicated Leaderboard tab.

Rule	Specification
Storage key	cabby_best_scores (JSON object mapping game ID to highest score number)
Update trigger	On every Save Score action on the Game page
Update condition	Update the value for the game ID only if the new score exceeds the stored best
Game page display	Shown above the iframe in the format "Your best: N", reading from cabby_best_scores for the current game ID
No-best fallback	When no best is recorded for the current game ID, display "Your best: none"
Leaderboard tab	A "Your Best" tab on the Leaderboard page
Your Best table	Columns: Rank, Game Name, Highest Score, Date achieved. One row per game with a recorded best
Sort order	Highest Score descending. Highest goes to Rank 1

8. Game data

Games are not built by the developer. They are defined as a static JavaScript array or JSON file. Each game object has the following fields.

Field	Type	Example
id	string	"speed-racer"
name	string	"Speed Racer"
genre	string	"racing"
rating	number	72
ageTag	string or null	"16+" or null
url	string	"https://example.com/game-embed"
description	string	"A fast-paced racing game"

The developer must include at least 9 game entries. Game URLs can be placeholder URLs during development as long as the iframe loading mechanism works correctly.

9. localStorage specification

All keys must be prefixed with "cabby_" to avoid conflicts with other sites.

Key	Value type	Description
cabby_username	string	Display name set by the user
cabby_scores	JSON array	Array of objects: {game, score, date}
cabby_favourites	JSON array	Array of game ID strings
cabby_missions_state	JSON object	{date: "YYYY-MM-DD", completed: [bool, bool, bool]}
cabby_streak	JSON object	{current, best, lastCompletedDate}
cabby_played_games	JSON array	Array of game ID strings the user has played
cabby_session_times	JSON object	Map of game ID to total seconds played
cabby_bookmarks	JSON array	Array of bookmark objects: {id, gameId, gameName, note, date, lastScore}
cabby_best_scores	JSON object	Map of game ID to the highest score recorded for that game

10. Visual design

The design must follow the layout shown in reference_image.png. The portal uses a dark theme throughout.

10.1 Colour palette

Colour	Hex code	Used for
Logo yellow	#F5C842	CABBY text, active sidebar icon, streak number, best score label
Primary green	#1D9E75	Action buttons, rating badges, avatar background, mission checkmark
Danger red	#E24B4A	Delete buttons, notification badge
Flame orange	#EF9F27	Streak flame icon in the navigation bar
Nav background	#2A2A2A	Top navigation bar, mission card surface
Sidebar background	#252525	Left sidebar
Page background	#1E1E1E	Main page background
Mission completed	#1D3328	Background for completed mission cards
Border colour	#333333	Borders and separators
Text primary	#EEEEEE	Headings and main text
Text secondary	#999999	Placeholder text, muted labels
Search bar bg	#3A3A3A	Search input and active sidebar icon background

10.2 Typography

Element	Size	Weight	Colour
Logo text	16px	500	#F5C842
Page headings	15px	500	#EEEEEE
Mission card title	12px	500	#DDDDDD
Game card name	12px	500	#DDDDDD
Rating badge	10px	500	#FFFFFF
Streak number	13px	500	#F5C842
Best score label	13px	500	#F5C842
Table text	13px	400	#CCCCCC
Stat card number	20px	500	#FFFFFF
Stat card label	12px	400	#999999

11. Technical requirements

Requirement	Specification
Tech stack	HTML, CSS, JavaScript. React is acceptable but not required
Backend	None
Database	None. No Firebase, Supabase, MongoDB, or any external data store
API keys	None. The project must not require any API keys to run
Data persistence	localStorage only. All keys prefixed "cabby_"
Game embedding	iframe with sandbox "allow-scripts allow-same-origin"
Routing	Hash-based or JS page switching
Page width range	Responsive across 768px to 1440px wide
Responsive at 1024px+	3-column game grid, full sidebar visible
Responsive at 768px to 1023px	2-column game grid, sidebar unchanged
Below 768px	Not required
Browser support	Latest Chrome, Firefox, Edge
Hosting	GitHub Pages or Vercel free tier. No paid hosting

12. Out of scope

Important

The following items are not part of this project. Do not build these. Do not include code for these. If you think any of these are needed, contact the client before proceeding.

- Building actual games from scratch
- Multiplayer or co-op features
- Chat or social features
- Payment processing or subscriptions
- Mobile app versions (iOS or Android)
- Admin dashboard or CMS
- Ads or monetisation
- Third-party authentication
- Any feature requiring API keys or a backend
- Adding new missions beyond the 10 templates
- Custom mission editing by users
- Support for screens below 768px

13. Acceptance criteria

These are the binary delivery-gate criteria. Every item must pass for the submission to be accepted. Granular per-feature scoring lives in the separate Rubric.

#	Criterion
1	[] All 4 pages exist and are reachable via navigation: Home, Game, Leaderboard, Profile.
2	[] Daily missions render exactly 3 cards per day, are deterministic by calendar date, auto-complete on the matching user action, and reset at local midnight.
3	[] Streak counter increments only when all 3 daily missions complete, persists across sessions, and resets correctly when a day is skipped.
4	[] Pause bookmarks can be saved from the Game page, list on the Profile page, and clicking a bookmark's game name navigates back to that Game page.
5	[] Best score tracker writes to <code>cabby_best_scores</code> , updates only when the new score is higher, and displays as <code>Your best: N</code> or <code>Your best: none</code> on the Game page.
6	[] All 9 documented <code>cabby_</code> -prefixed <code>localStorage</code> keys are implemented with the exact schemas specified in the Data Persistence section.
7	[] No backend, database, third-party auth, API keys, or paid services are present in the codebase or at runtime.
8	[] Game embedding uses an <code><iframe></code> with <code>sandbox="allow-scripts allow-same-origin"</code> and shows the documented error fallback message when the game URL fails to load.
9	[] Visual implementation honors the documented colour palette (12 hex values), typography table, and layout dimensions (page width 768-1440px, sidebar 48px, game card 16:10 aspect ratio with 8px border radius).
10	[] Project is deployed to GitHub Pages or Vercel free tier with a working public URL, and the repository contains a <code>README.md</code> at the root.
11	[] No placeholder text appears in the final delivery: no Lorem ipsum, no TODO, no FIXME, and no sample-text artifacts in user-visible UI or game data.

All 11 items must pass for the submission to be accepted. There is no partial-credit threshold at this gate. Granular per-feature scoring is applied separately to the accepted build and lives in the project's rubric file. Up to 2 rounds of revisions after delivery. Changes outside this PRD require a separate agreement.

14. Developer choices and accepted options

The following decisions are left to the freelancer's judgement. Any option in each row is acceptable.

Decision	Accepted options
Framework	Vanilla JS, React, or any static-only stack
Routing approach	Hash-based routing, single-page JS switching, multi-page navigation
Build tool	None, Vite, Create React App, any free static builder
Hosting platform	GitHub Pages, Vercel free tier
Icon library	Any free icon library (Lucide, Tabler, Heroicons), inline SVG
Genre gradient palettes	Developer chooses tones, must remain dark, genre-suggestive

Decision	Accepted options
Game data source	Static array in JS file, static JSON file in repo
Mission selection formula	Any deterministic function of date that yields 3 distinct missions from the pool of 10
Bookmark ID generation	Any unique string approach (timestamp, UUID, random hash)

15. Tool requirements

Category	Specification
Required tools	Any code editor that saves UTF-8 plain text (VS Code, Sublime, Notepad++, etc.), Git
Acceptable alternatives	Any text editor that can save UTF-8 plain text
Forbidden	No-code site builders, AI-generated full-site code generators that produce unreadable output

16. Build and export specifications

Requirement	Specification
Build output	The live site must be deployable as static files only. No server runtime required
Source files	All source code committed to the GitHub repository in raw, editable form
Production build	If using React, the production build output must also be deployable on a static host

End of document